

LESSON 8-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Equations and Solutions**

Maryland Standard 6.EE.B.5 • Learning target – I can check whether a value is a solution of an equation by substituting it.

REMEMBER

- A **solution** is a value of the variable that makes an equation **true**.
- To check a value, **substitute** it for the variable. Then compare the two sides.
- If the two sides are not equal, the value is **not** a solution.

WORKED EXAMPLEIs $n = 14$ a solution of $n + 9 = 23$?**Step 1** Replace n with 14: $14 + 9 = 23$ **Step 2** Compare: $23 = 23$. Both sides match.**Answer Yes, $n = 14$ is a solution.****WORD BANK****equation** – two amounts are equal**solution** – the number that makes it true**substitute** – put the number in for the letter**4p** – 4 times p**PRACTICE***show your work • label units*

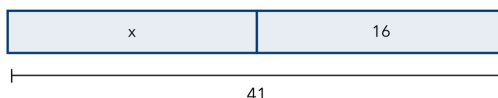
1. Substitute the value. Write **T** for true or **F** for false.

a. $y + 8 = 20$ ($y = 12$) ____ b. $5m = 35$ ($m = 6$) ____ c. $k - 9 = 16$ ($k = 25$) ____

d. $h \div 4 = 7$ ($h = 32$) ____ e. $13 + w = 30$ ($w = 17$) ____ f. $6p = 54$ ($p = 9$) ____

g. $q - 11 = 4$ ($q = 15$) ____ h. $9r = 63$ ($r = 8$) ____

2. The tape diagram shows $x + 16 = 41$. Which value is the solution: 15, 25, or 35? Substitute each value to check.



Solution _____

3. **Bike Rack.** A bike rack held b bikes. 7 bikes **left**. 18 bikes were still there. Kendra writes $b - 7 = 18$. Is $b = 11$ a solution? Is $b = 25$ a solution? Show both checks.

Show your work.

Solution _____

- 4. Find the Error.** Andre says $p = 8$ is a solution of $4p = 12$ because $4 + 8 = 12$. Explain his mistake. Then find the real solution.

Frame Andre is wrong because $4p$ means _____. The solution is $p =$ _____.

- 5. Stretch.** Malik paid \$42 for t movie tickets that cost \$6 each, so $6t = 42$. Is $t = 36$ a solution? Is $t = 7$? Explain which value makes sense.

Step 1 Put 36 in for t . Does 6×36 equal 42?

Step 2 Put 7 in for t . Check again.

Solution _____

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Make up an equation for someone at home. Check their guess together by substituting.

LESSON 8-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Writing Equations**

Maryland Standard 6.EE.B.6 • Learning target – I can write an equation for a situation and test a proposed answer.

REMEMBER

- Use a **variable** for the unknown amount.
- “7 less than n ” is $n - 7$, **not** $7 - n$.
- **Test** an answer by substituting it.

WORKED EXAMPLE b books plus 15 more is 52 books. Test $b = 37$.**Step 1** Equation: $b + 15 = 52$ **Step 2** Test: $37 + 15 = 52$, so it checks.**Answer** $b + 15 = 52$; $b = 37$ **WORD BANK****increased by** – add (+)**less than** – subtract (–)**times** – multiply ($\times$)**divided by** – divide ($\div$)**PRACTICE***show your work • label units*

1. Write an equation for each sentence. Use
- n
- for the number.

Step 1 Find the key word.**Step 2** Write n , the sign, the number, = and the total.**a.** A number increased by 9 is 31. _____ **b.** 7 times a number is 63. _____**c.** 12 less than a number is 20. _____ **d.** A number divided by 5 is 8. _____

- 2.
- Stickers.**
- Rosa gave away 14 stickers and has 29 left. Write an equation for
- s
- . Test
- $s = 43$
- .

**Equation and test** _____

- 3.
- Field Trip.**
- 56 students ride in 4 vans. Each van has the
- same**
- number. Write an equation for
- v
- , the students in each van. Test
- $v = 12$
- and
- $v = 14$
- .

Step 1 Vans $\times$ students in each van = all students.**Step 2** Put 12 in for v . Then try 14.*Show your work.***Equation and solution** _____

4. **Reading Log.** Luis reads the same number of pages, p , each day for 5 days. He reads 85 pages in all. Write an equation. Test $p = 17$.

Step 1 Pages each day $\times$ 5 days = 85 pages.

Step 2 Put 17 in for p and check.

Equation and test _____

5. **Find the Error.** Tia is 13. She is 4 years older than her brother. Omar writes $b - 4 = 13$ for his age. Explain the mistake. Find b .

Frame Omar is wrong because the brother is ____ than Tia. The equation should be ____, so $b =$ ____.

TEST PRACTICE

choose the best answer

6. Jada saved \$9 a week for w weeks. She saved \$72. Which equation fits?

A $w + 9 = 72$

B $9w = 72$

C $w \div 9 = 72$

D $72w = 9$

7. Which value of n is the solution of $n - 15 = 27$?

A $n = 12$

B $n = 27$

C $n = 42$

D $n = 405$

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____